



UT Dallas 2026 Undergraduate Catalog

# School of Natural Sciences and Mathematics

## Data Science (BS)

The Data Science BS is jointly offered by the Department of Mathematical Sciences in the School of Natural Sciences and Mathematics and the Department of Computer Science in the Erik Jonsson School of Engineering and Computer Science. Data Science is an emerging discipline that lies at the intersection of Computer Science, Mathematics, and Statistics. The curriculum for the program provides a solid foundation in all the three disciplines. It prepares students for Data Scientist or related positions in industry, business, and government that are currently in high demand and also for graduate study in either of the three disciplines. The curriculum also includes a capstone project course.

## Bachelor of Science in Data Science

*Degree Requirements (120 semester credit hours)*

*[View an Example of Degree Requirements by Semester](#)*

## NSM Faculty

**Professors:** Swati Biswas, Min Chen, Pankaj Choudhary, Baris Coskunuzer, Mieczyslaw Dabkowski, Vladimir Dragovic, Sam Efromovich, Wieslaw Krawcewicz, Susan Minkoff, L. Felipe Pereira, Dmitry Rachinskiy, Viswanath Ramakrishna, Janos Turi

**Associate Professors:** Maxim Arnold, Yan Cao, Liang Hong, Oleg Makarenkov, Tomoki Ohsawa, Anh Tran

**Assistant Professors:** Carlos Arreche, Noirrit Chandra, Ronan Conlon, Rizwanur Khan, Qiwei Li, Stephen McKeown, Chuan-Fa Tang, Jiayi Wang, Nathan Williams, Nan Wu

**Professors Emeriti:** Larry Ammann, Ali Hooshyar, Patrick Odell, John Van Ness

**Clinical Professor:** Natalia Humphreys

**Clinical Associate Professor:** Mohammad Akbar

**Clinical Assistant Professor:** Wenyi Lu

**Professors of Instruction:** Anatoly Eydelzon, Manjula Foley, Bentley Garrett, Yuly Koshevnik

**Associate Professors of Instruction:** Mohammad Ahsan, Kelly Aman, Malgorzata Dabkowski, Rabin Dahal, Derege Mussa, My Linh Nguyen, Jigarkumar Patel, Julie Sutton

**Assistant Professors of Instruction:** Anani Komla Adabrah, Iris Alvarado, Saikat Biswas, Hui Ding, Kemelli Estacio-Hiroms, Huizhen Guo, Shengjie Jiang, Joselle Kehoe, Runzhou Liu, Neha Makhijani, Irina Martynova, Diarisoa Mihaja Rakotomalala, Adrian Murza, Ajaya Paudel, Octavious Smiley, Nasrin Sultana, Che-Yu Wu

## ECS Faculty

**Professors:** Farokh B. Bastani, Sergey Bereg, Ovidiu Daescu, Yvo G. Desmedt, Ding-Zhu Du, Paul Fishwick, Xiaohu Guo, Gopal Gupta, Kevin Hamlen, Sanda M. Harabagiu, Dung T. Huynh, Jason Jue, Latifur Khan, Andrian Marcus, Neeraj Mittal, Dan I. Moldovan, Sriraam Natarajan, Yu-Chung (Vincent) Ng, Tien Nguyen, Simeon C. Ntafos, Ravi Prakash, Kamil Sarac, Haim Schweitzer, Bhavani Thuraisingham, W. Eric Wong, Weili Wu, I-Ling Yen, Rym Zalila-Wenkstern

**Associate Professors:** Feng Chen, Lawrence Chung, Jorge A. Cobb, Vibhav Gogate, Benjamin Raichel, Nicholas Ruozzi

**Assistant Professors:** Xinya Du, Emily Kyle Fox, Yunhui Guo, Shuang Hao, Rishabh Iyer, Kangkook Jee, Chung Hwan Kim, Jin Kim, Jessica Ouyang, Yapeng Tian, Shiyi Wei, Yu Xiang, Wei Yang

**Professors Emeriti:** R. Chandrasekaran, Zygmunt Haas, Ivor P. Page, William J. Pervin, Balaji Raghavachari, Ivan Hal Sudborough, Subbarayan Venkatesan, Kang Zhang

**Professors of Instruction:** Ebru Cankaya, John Cole, Timothy (Tim) Farage, Shyam Karrah, Pushpa Kumar, Nhut Nguyen, Greg Ozbirn, Miguel Razo-Razo, Jeyakesavan (Jey) Veerasamy

**Associate Professors of Instruction:** Sridhar Alagar, Gordon Arnold, Anjum Chida, Wei Pang Chin, Bhadrachalam Chitturi, Chris I. Davis, Karen Doore, Neeraj Gupta, Khiem Le, Anarag Nagar, Mehra Nouroz Borazjany, Jalal Omer, Mark Paulk, Jason W. Smith, Laurie Thompson, James Willson, Nurcan Yuruk

**Assistant Professors of Instruction:** Eric Becker, Scott Dollinger, Serdar Erbatur, Ranran Feng, Omar Hamdy, Gity Karami, Kamran Khan, Karen Mazidi, Richard K. Min, Priya Narayanasami, Brian Ricks, Elmer Salazar, Meghana Satpute, Nidhiben Solanki, Srimathi Srinivasan, Yi Zhao

## I. Core Curriculum Requirements: 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

### Communication: 6 semester credit hours

Select any 6 semester credit hours from **Communication Core** courses (see advisor)

### Mathematics: 3 semester credit hours

**MATH 2417** Calculus I<sup>1, 2</sup>

Or select any 3 semester credit hours from **Mathematics Core** courses (see advisor)

**Life and Physical Sciences: 6 semester credit hours**<sup>1, 3</sup>**PHYS 2325** Mechanics

or **PHYS 2421** Honors Physics I - Mechanics and Heat

**PHYS 2326** Electromagnetism and Waves

or **PHYS 2422** Honors Physics II - Electromagnetism and Waves

Or select any 6 semester credit hours from **Life and Physical Sciences Core** courses (see advisor)

**Language, Philosophy and Culture: 3 semester credit hours**

Select any 3 semester credit hours from **Language, Philosophy and Culture Core** courses (see advisor)

**Creative Arts: 3 semester credit hours**

Select any 3 semester credit hours from **Creative Arts Core** courses (see advisor)

**American History: 6 semester credit hours**

Select any 6 semester credit hours from **American History Core** courses (see advisor)

**Government/Political Science: 6 semester credit hours**

Select any 6 semester credit hours from **Government/Political Science Core** courses (see advisor)

**Social and Behavioral Sciences: 3 semester credit hours**

Select any 3 semester credit hours from **Social and Behavioral Sciences Core** courses (see advisor)

**Component Area Option: 6 semester credit hours****MATH 2417** Calculus I<sup>1, 2</sup>**MATH 2419** Calculus II<sup>1, 2</sup>**PHYS 2125** Physics Laboratory I<sup>1, 3</sup>

or **PHYS 2421** Honors Physics I - Mechanics and Heat<sup>4, 5</sup>

Or select any 6 semester credit hours from **Component Area Option** courses (see advisor)

## II. Major Requirements: 65-66 semester credit hours

**Major Preparatory Courses: 19-20 semester credit hours beyond Core Curriculum****CS 1436** Programming Fundamentals**CS 1337** Computer Science I**CS 2336** Computer Science II

**MATH 3315** Discrete Mathematics and Combinatorics

or **CS 2305** Discrete Mathematics for Computing I<sup>6</sup>

**MATH 2417** Calculus I<sup>1, 2, 7</sup>

**MATH 2418** Linear Algebra

**MATH 2419** Calculus II<sup>1, 2, 7</sup>

**NATS 1101** Natural Sciences and Mathematics Freshman Seminar

**PHYS 2325** Mechanics<sup>1, 3</sup>

and **PHYS 2125** Physics Laboratory I<sup>1, 3</sup>

or **PHYS 2421** Honors Physics I - Mechanics and Heat<sup>1, 3, 8</sup>

**PHYS 2326** Electromagnetism and Waves<sup>1, 3</sup>

or **PHYS 2422** Honors Physics II - Electromagnetism and Waves<sup>1, 3</sup>

**PHYS 2126** Physics Laboratory II

### Major Core Courses: 46 semester credit hours

**CS 3345** Data Structures and Foundations of Algorithmic Analysis

**CS 4347** Database Systems

**CS 4371** Foundations of Big Data Management and Analytics

**CS 4372** Computational Methods for Data Scientists

**CS 4375** Foundations of Machine Learning

**MATH 3310** Theoretical Concepts of Calculus

**MATH 3351** Advanced Calculus

**MATH 4301** Mathematical Analysis I

**STAT 3355** Introduction to Data Analysis

**STAT 4351** Probability

**STAT 4352** Mathematical Statistics

**STAT 4354** Numerical and Statistical Computing

**STAT 4355** Applied Linear Models

**STAT 4360** Introduction to Statistical Learning

**STAT 4475** Capstone Project

or **CS 4475** Capstone Project

or **MATH 4475** Capstone Project

## III. Elective Requirements: 12-13 semester credit hours

### Guided Electives: 12-13 semester credit hours

Although both lower- and upper-division courses may count as guided electives, they must be approved by the advisor.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

**NOTE:** Students transferring into this program at the upper-division level are expected to have completed all of the 1000- and 2000- level Mathematics and Computer Science core course requirements.

## Fast Track Baccalaureate/Master's Degrees

For students with strong academic record who intend to pursue master's studies at UT Dallas, accelerated BS/MS Fast Tracks are available. After Fast Track admission to an MS program, students may take up to 15 semester credit hours of approved graduate courses in their senior year to use toward completion of both the BS and MS degrees. The available Fast Track options and their admission requirements, in addition to having completed 90 or more semester credit hours (out of which 36 hours are from the core curriculum) with a cumulative GPA of at least 3.00, are as follows:

- MS in Mathematics (Data Science specialization and possibly other specializations<sup>9</sup>): Students must have a cumulative GPA of at least 3.20 in all mathematics and statistics courses.
- MS in Statistics: Students must have a cumulative GPA of at least 3.20 in all mathematics and statistics courses.

Interested students, after reviewing the [UT Dallas Fast Track policy](#), should contact their undergraduate advisor and the graduate advisor of the intended MS program well in advance of their junior year to prepare a course sequence permitting maximal advantage and apply to the Fast Track program.

## Certificates

### Certificate in Data Science

*15 semester credit hours*

A Certificate in Data Science is offered by the Department of Mathematical Sciences in the School of Natural Sciences and Mathematics.

The focus of the Certificate in Data Science is to provide training in core data analytics skills, including programming and statistical and machine learning methods.

### Admission Requirements

Two semesters of Calculus.

# Certificate Requirements

Students must complete the following courses:

**MATH 2333** Matrices, Vectors, and Data

**MATH 4332** Scientific Computing using Python

**MATH 4355** Methods of Applied Mathematics

**STAT 3355** Introduction to Data Analysis

**STAT 4360** Introduction to Statistical Learning

1. A required Major course that also fulfills Core Curriculum requirements. If semester credit hours are counted in the Core Curriculum, students must complete additional coursework to meet the minimum requirement for graduation. Course selection assistance is available from the undergraduate advisor.
2. Three semester credit hours of Calculus are counted to fulfill the Mathematics Core Requirement with the remaining one semester credit hour to be counted under Component Area Option Core.
3. Six semester credit hours of Physics are counted under Science core, and one semester credit hour of Physics (PHYS 2125) is counted under Component Area Core.
4. Please consult your advisor if selecting Honors Physics.
5. Students may use three semester credit hours of PHYS 2421 to count under Science core, and one semester credit hour of PHYS 2421 under Component Area Option core.
6. BS in Data Science students can substitute MATH 2312 with MATH 2413.
7. MATH 2417 and MATH 2419 requirements can be fulfilled by completing MATH 2413, MATH 2414, and MATH 2415.
8. Students who complete PHYS 2421 do not need to complete PHYS 2125.
9. For other specializations, please consult with the Graduate Advisor for Mathematics regarding admission prerequisites.

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